

Multiple Choice (10 marks)

1. If $x - 3 = 6$, what is the value of x ?
 - (a) 0
 - (b) 4
 - (c) 3
 - (d) 2
 - (e) 9
2. Let $f(x,y) = x^2 - 2xy + y^3$ for all real x and y . Which of the following is true?
 - (a) f has all of its relative extrema on the line $y = x^2$.
 - (b) f has all of its relative extrema on the line $x = y$.
 - (c) f has all of its relative extrema on the line $x = 2y$.
 - (d) f has a relative maximum at $(0, 0)$.
 - (e) f has an absolute maximum at $(2/3, 2/3)$.
3. For the regression line, which of the following statements about residuals is true?
 - (a) The sum of the residuals is always one.
 - (b) If the correlation is 0, there will be a distinct pattern in the residual plot.
 - (c) The larger the residuals, the better the linear model.
 - (d) The residuals are always negative.
 - (e) If the correlation is 1, there will be a distinct pattern in the residual plot.
4. The polynomial $x^3 - Ax + 15$ has three real roots. Two of these roots sum to 5. What is $|A|$?
 - (a) 19.0
 - (b) 17.0
 - (c) 25.0
 - (d) 23.0
 - (e) 18.0
5. Find $13 \text{ over } 14 + 7 \text{ over } 14$.
 - (a) $20 \text{ over } 28$
 - (b) $2 \text{ and } 3 \text{ over } 14$
 - (c) $1 \text{ and } 3 \text{ over } 7$
 - (d) $2 \text{ and } 1 \text{ over } 14$
 - (e) $3 \text{ and } 2 \text{ over } 7$

True / False (10 marks)

Answer True or False

6. True or False: The opposite of the number -7 is 7, which represents its additive inverse on the number line.
7. True or False: In a set of 20 positive integers, there are at least 10 pairs of numbers that have a difference which is a multiple of 10.
8. True or False: Morse code can represent 30 distinct symbols using sequences of 1, 2, 3, or 4 total dots and dashes.
9. True or False: A correlation of 0.6 explains four times the percentage of variation in y compared to a correlation of 0.3.
10. True or False: The set $\{2, 3, 4\}$ contains values that satisfy the inequality $2x - 1 < 10$ for all elements in the set.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. The graph of $xy = 1$ is a hyperbola. Find the distance between the foci of this hyperbola.
12. Calculate the determinant of the matrix $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 7 & 0 & 0 & 0 & 0 \\ 3 & 8 & 6 & 0 & 0 & 0 \\ 4 & 9 & 5 & 2 & 1 & 4 \\ 5 & 8 & 4 & 0 & 2 & 5 \\ 6 & 7 & 3 & 0 & 3 & 6 \end{bmatrix}$ and discuss the methods used in your calculation.

End of Paper